

4. Library Book Reservation System for SIMATS Library using SaaS

Aim

To develop a **Cloud-Based Library Book Reservation System for the SIMATS Library** using a Cloud Service Provider (such as **Zoho Creator, Google AppSheet, Microsoft Power Apps, or Salesforce**) to demonstrate the **Software as a Service (SaaS)** model. The application enables students to search, reserve, and borrow library books while allowing librarians to manage book inventory and reservation records through a cloud-based platform.

Procedure

1. Sign in to a cloud platform (e.g., Zoho Creator).
2. Create a new application named **SIMATS Library Book Reservation System**.
3. Design forms for Students, Books, and Reservations.
4. Add book details such as title, author, ISBN, category, and availability.
5. Create a student registration form.
6. Develop a reservation form for booking library books.
7. Configure workflows to automatically generate Reservation IDs and update book availability.
8. Create reports for available books, reserved books, issued books, and reservation history.
9. Test the application using sample student and book records.
10. Deploy the application on the cloud for web-based access.

Implementation

Step 1: Create the Application

- Log in to the cloud platform.
- Create a new application named **SIMATS Library Book Reservation System**.

Step 2: Create Forms

Student Form

Fields:

- Student ID
- Student Name
- Department
- Register Number
- Email
- Mobile Number

Book Form

Fields:

- Book ID
- Book Title
- Author
- ISBN
- Category
- Number of Copies
- Availability Status

Reservation Form

Fields:

- Reservation ID
 - Student Name (Lookup)
 - Book Title (Lookup)
 - Reservation Date
 - Issue Date
 - Return Date
 - Reservation Status
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Step 3: Add Automation

- Automatically generate Reservation IDs.
 - Update the book status to **Reserved** or **Issued**.
 - Mark the book as **Available** after it is returned.
 - Display reservation confirmation.
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Step 4: Create Reports

Generate reports for:

- Available Books
 - Reserved Books
 - Issued Books
 - Student Details
 - Reservation History
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Step 5: Test the Application

- Add sample books.
 - Register students.
 - Reserve books.
 - Verify availability updates correctly.
 - View generated reports.
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Result

A **Cloud-Based Library Book Reservation System for the SIMATS Library** was successfully developed using a **Software as a Service (SaaS)** cloud platform. The application enables students to reserve books online, helps librarians manage inventory efficiently, and provides centralized cloud storage, automated workflows, and real-time reporting.